BME 6938 · ADVANCED BIOMEDICAL AI · SPRING 2027

LECTURE 13-1

Deployment and Human-AI Teaming

What happens after the model is good enough



Parisa Rashidi · Herbert Wertheim College of Engineering, University of Florida

Where We Are

The last content week: a model meets a hospital.

Week 12	Week 13: today	Weeks 14-15
Evaluation, robustness, fairness	**Deployment and human-AI teaming**	**Project presentations**
What a score does and does not tell you	What happens after the model is good enough	No lectures or paper sessions

Four Things Training Does Not Decide

Every one of them changes what the model is worth.

Where it is validated	performance at your hospital, not the vendor's
What threshold it runs at	how many patients get reviewed per true case
Whether anyone is watching	the inputs change after go-live
What the clinician does with it	the combination has to be measured



What External Validation Found

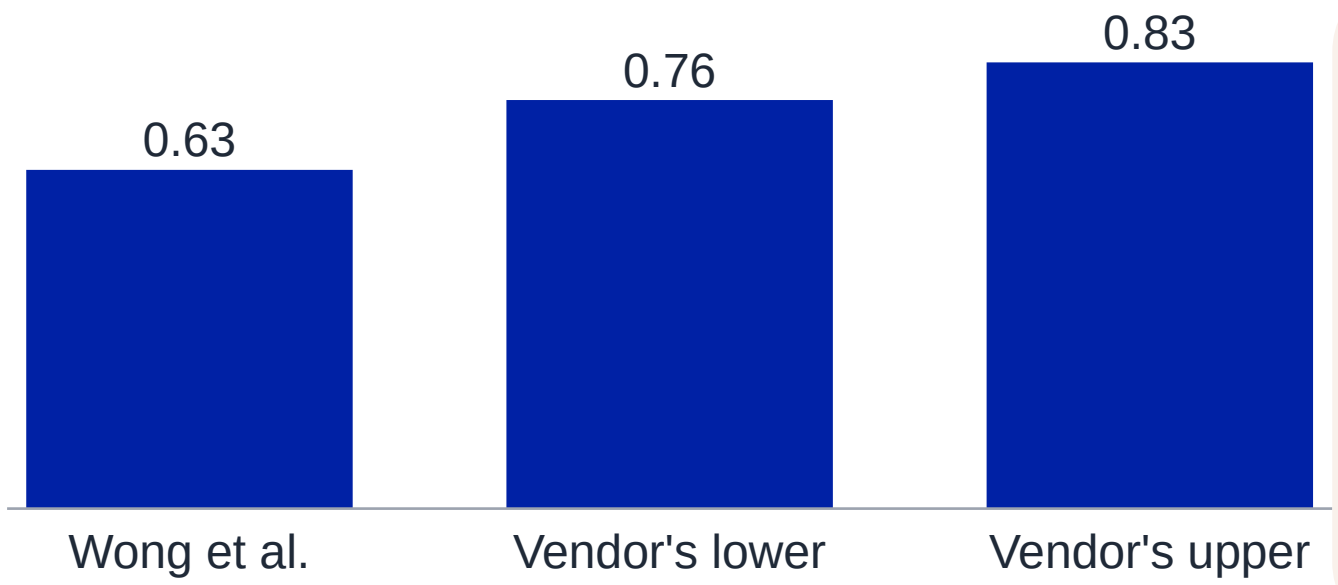
The Model and the Study

A proprietary sepsis model, scored every 15 minutes, validated at one health system.

The model	proprietary, deployed at hundreds of hospitals, scored every 15 minutes
The cohort	27,697 patients, 38,455 hospitalizations, Dec 2018 to Oct 2019
The outcome	sepsis in 2,552 hospitalizations: 7% prevalence
The comparison	the vendor's own reported performance, from internal documentation

Discrimination

AUC 0.63 at this health system, against the 0.76 to 0.83 the vendor reported.



area under the ROC curve

The objection, answered

per-horizon AUCs are 0.72 to 0.76, and the authors call them misleading: the predictions are not independent

calibration was poor throughout

At the Recommended Threshold

Two thirds of sepsis missed, and alerts on 18% of hospitalizations.

Sensitivity	33%
Specificity	83%
Positive predictive value	12%
Hospitalizations alerted	18% (6,971 of 38,455)
Number needed to evaluate	8

Note

the article carries a 2 August 2021 correction to this statistic

which is the process working

The Question That Actually Matters

Not what the model scored, but what the alert added.

2,552	sepsis cases
1,709	never flagged (67%)
1,030	of those got timely antibiotics anyway
183	flagged before antibiotics were given in time (7%)

Median lead time where it did fire: 2.5 hours (IQR 0.5 to 15.6).

Animation: What External Validation Found

Animation clip: week13-clip1.mp4

Silent version embedded (click to play). The narrated version is in the course repository.

What to Take From It

Two rules students should carry into their own projects.

Validate locally

your population, your pipeline

case mix, coding, and ordering practice all differ

Measure what it adds

compare against current practice, not against nothing

most studies never make that comparison



Thresholds, Alert Burden, and Drift

One Model, Three Operating Points

At 7% prevalence, the threshold decides the alert load.

Sensitivity	PPV	Alerts per 100 admissions	Patients reviewed per true case
80%	10%	56.0	10
50%	20%	17.5	5
25%	35%	5.0	2.9

The sepsis model at its recommended threshold sits near the first row: 8 reviewed per true case.

Illustrative operating points at 7% prevalence, as in clip 2.

Rarity Governs the Alert Burden

A much better model lands in the same place.

	Deployed model (Wong et al.)	Notebook model
AUC	0.63	0.78
Sensitivity at the chosen point	33%	82%
Positive predictive value	12%	13%
Patients reviewed per true case	8	8.0

A 15-point gain in AUC bought no reduction in the review burden.

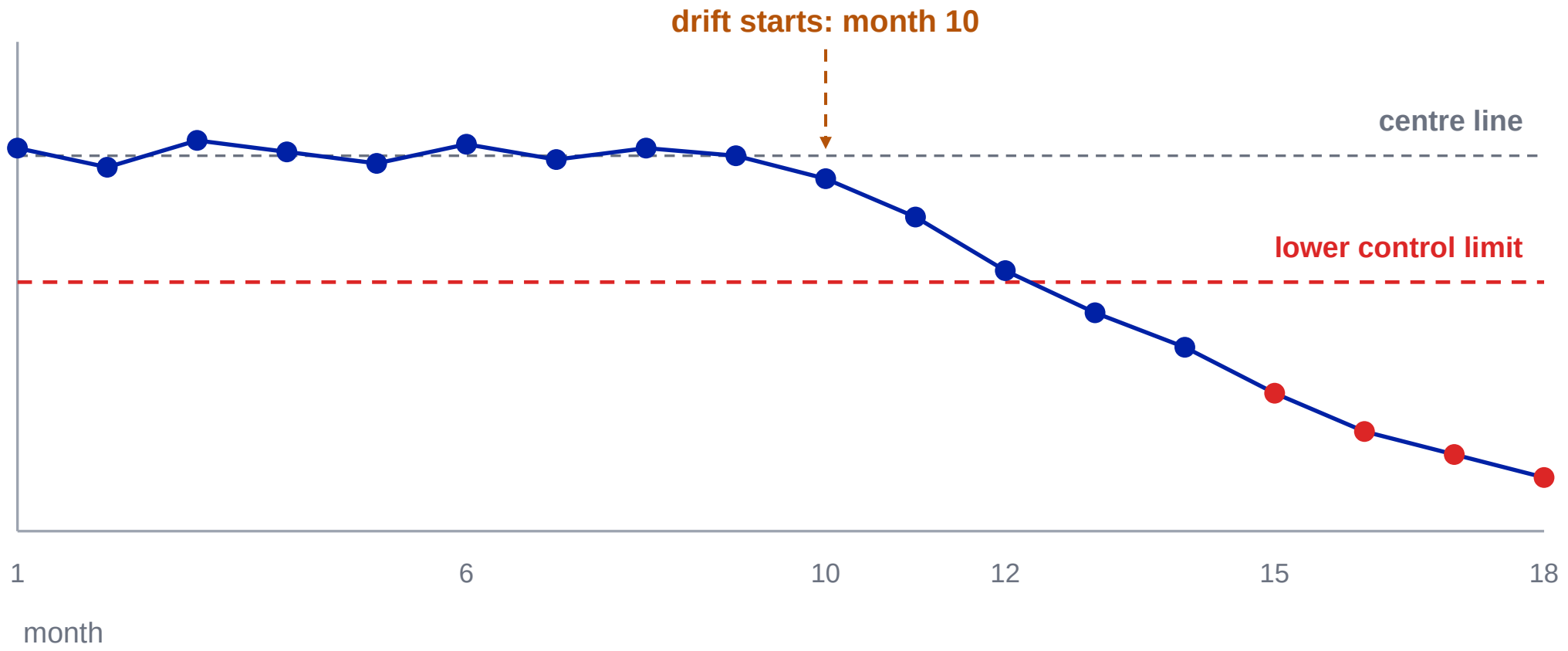
Nothing Holds Still

A new assay, a coding change, a different case mix.

Inputs	a new lab assay, a changed unit, a renamed field
Coding	documentation and billing practice change
Population	a new service line, a new season, a new referral pattern
The model	keeps producing scores, with no error of any kind

Watch It: A Control Chart

Monthly performance against limits set from a stable period.



Monitoring Buys a Dated Signal, Not a Solution

Three-sigma fires in month 15, five months late.

Drift starts	month 10
Run-of-three rule fires	month 12, two months of exposure
Three-sigma rule fires	month 15, five months of exposure

Write the monitoring plan before go-live: metrics, limits, who looks, and what happens when a limit is crossed.

Animation: Thresholds, Alert Burden, and Drift

Animation clip: week13-clip2.mp4

Silent version embedded (click to play). The narrated version is in the course repository.



Humans and Models Together

Two Obvious Rules

Flag when either flags, or flag only when both do.

Flag if either flags

catches more

alarms more

the usual choice when a miss

is worse than a false alarm

Flag only if both flag

alarms less

misses more

the usual choice when alert

fatigue is the problem

Clinician sensitivity 0.80, model 0.75 (illustrative).

Same Two Sensitivities, Opposite Answers

What changes is whether their mistakes coincide.

	Errors independent	Misses coincide
Flag if either flags	0.95	0.80 (the clinician alone)
Flag only if both flag	0.60	0.75 (the model alone)

The whole result lives in a quantity you cannot compute from the two sensitivities.

The notebook sweeps the correlation: either-flags falls from 0.951 to 0.800, both-flag rises from 0.603 to 0.750.

And People Are Not Rules

Automation bias, alert fatigue, and a trial that found no benefit.

Automation bias	a confident output is trusted more than it has earned
Alert fatigue	alerts that are usually wrong stop being read at all
Measured, not assumed	Goh et al. (Week 12) found no benefit from the combination

Ambient Intelligence

Sensing the care space instead of waiting to be told.

Hand hygiene (Singh et al., 2020)

16 depth sensors, acute care unit

algorithm: 92.1% sensitive

human observers: 85.2%

test set: 88 dispenser-use events

ICU mobility (Yeung et al., 2019)

7 depth sensors, one hospital

563 mobility events

activity detected: 87.2% sensitivity

staff counted correctly: 68.8%

The weak numbers are the teaching: 88 events, and a 69% count.

88 Events Is Not Many

The bootstrap from Week 12, applied to the hand-hygiene claim.



Difference: 0.000 to 0.136. Ahead in 96.7% of resamples.

Animation: Humans and Models Together

Animation clip: week13-clip3.mp4

Silent version embedded (click to play). The narrated version is in the course repository.

Governance Before Technique

Sensing a care space is a consent question first.

Who is observed	patients and staff, and staff are rarely asked
What is kept	depth rather than images helps, and is not the whole answer
What it is used for	a record collected for care can be read for other purposes
What the review names	clinical validation, data privacy, model transparency

Recap: After the Model Is Good Enough

Validate	Locally, and against current practice. AUC 0.63, and 7% of cases helped.
Threshold	Rarity governs the burden: AUC 0.78 still gave 8 reviewed per case.
Monitor	A drift from month 10 was caught in month 15. Write the plan first.
Team	0.95 or 0.80 from the same two sensitivities. Measure the pair.

Before Thursday, and Then the Projects

Read	Wong et al., External validation of a sepsis model, and Haque et al., Ambient intelligence. The last paper sessions.
Write	Reading responses on Canvas before 1:00 PM Thursday.
Explore	Week 13 reference notebook (ungraded): the threshold sweep, a drift nobody catches in time, teaming, and 88 events.
Then	Presentations Apr 13, 15 and 20: every team's slides are due 1:00 PM Apr 13. Paper and repository due Apr 21, 11:59 PM.